Analytical Chemistry Cumulative Exam
December 2, 2011

Basic Electrochemistry

(At the end of this cumulative exam are equations and constants and tables that might be helpful in answering these questions)

1. (15 points) Define what a "reference electrode" is in the electrochemistry field. What are the desirable features that a reference electrode should possess? Give an example of a reference electrode and its half-reaction. Draw a diagram to show what it might look like.

2. (10 points) What is the standard electrode potential for the reduction of Ni^{2+} to Ni° vs. the Ag/AgCl (saturated KCl) reference electrode?

(Note: for $AgCl_{(s)} + e = Ag_{(s)} + Cl^-$, for saturated KCl, E° = +0.197 V vs. NHE)

3. Several hydrocarbons and carbon monoxide have been studied as possible fuels for use in fuel cells. The E° values for the half reactions at 25 °C are shown below.

	E^0 (V vs. NHE)
$CO_{(g)} + H_2O_{(l)} = CO_{2(g)} + 2H^+ + 2e$	-0.103
$CH_{4(g)} + 2H_2O_{(l)} = CO_{2(g)} + 8H^+ + 8e$	0.169
$C_2H_{6(g)} + 4H_2O_{(l)} = 2CO_{2(g)} + 14H^+ + 14e$	0.143
$C_2H_{2(g)} + 4H_2O_{(l)} = 2CO_{2(g)} + 10H^+ + 10e$	-0.051

(a) (20 points) What is the oxidation state of carbon in the reactants and products of each of these half-reactions?

(b) (25 points) Calculate the cell voltage when the CO reaction is coupled with the standard oxygen half-cell in acid solution. (Use standard conditions, and assume activities are unity.)

(c) (10 points) Using the value that you obtained in (b) above, calculate the net work per mole of CO oxidized.

(d) (10 points) Using the value that you obtained in (b) or (c) above, calculate the net work per gram for CO oxidized. (Note that the molar mass for CO is 28.01 g/mol).

(e) (10 points) Here are the cell voltage, work per mole, and work per gram for CH_4:
1.060 V vs. NHE, -818.2 kJ/mol, and -51.0 kJ/g, respectively.
(Note that the molar mass for CH_4 is 16.04 g/mol).

Why do you think that although CO has an overall larger cell voltage than CH_4, CO offers far worse work per mole and work per gram than CH_4?

Constants, Equations, Conversion Factors, and Tables of Data

Ideal Gas Constant and Faraday Constant

$R = 0.08205746(14)\ \underline{L}\ \underline{atm}\ K^{-1}\ mol^{-1} = 8.3144621(75)\ \underline{J}\ \underline{K}^{-1}\ \underline{mol}^{-1}$

$F = 96485.3$ Coul/mol e^-

$RT/F = 0.02569$ V at 25 °C

$f = F/RT = 38.92\ V^{-1}$ at 25 °C

Temperature

0 °C = 273.15 K

Δ°C = ΔK

Room temperature = 25 °C = 273.15 + 25 = 298.15 K

Energy

1 J = 1 kg (m^2/s^2)=1 Coul V

1.000 kWh = 3.6000×10^6 J

1 cal = 4.184 J

1 eV = 1.60217653 10^{-19} J

1 kWh = 3.6×10^6 J

1 Wh = 3600 J

$$\Delta G = \Delta G^o + RT \ln \prod a^{\nu_i}$$

$$\Delta G^o_{rxn} = \sum \Delta G^o_{f,products} - \sum \Delta G^o_{f,\text{reactants}}$$

$$\Delta G_{rxn} = -n\ F\ E_{rxn} \quad (\textit{note}:\ \Delta G^o = -nFE^o)$$

$$\Delta E_{rxn} = E_{right} - E_{left} = E_{cathode} - E_{anode}$$

$$E = E^\circ + \frac{RT}{nF} \ln \prod a^{\nu_i} = E^\circ + \frac{RT}{nF} \ln \frac{(O)}{(R)}$$

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APPENDIX

C

REFERENCE TABLES

TABLE C.1 Selected Standard Electrode Potentials in Aqueous Solutions at 25°C in V *vs.* NHE[a]

Reaction	Potential, V
$Ag^+ + e \rightleftharpoons Ag$	0.7991
$AgBr + e \rightleftharpoons Ag + Br^-$	0.0711
$AgCl + e \rightleftharpoons Ag + Cl^-$	0.2223
$AgI + e \rightleftharpoons Ag + I^-$	−0.1522
$Ag_2O + H_2O + 2e \rightleftharpoons 2Ag + 2OH^-$	0.342
$Al^{3+} + 3e \rightleftharpoons Al$	−1.676
$Au^+ + e \rightleftharpoons Au$	1.83
$Au^{3+} + 2e \rightleftharpoons Au^+$	1.36
p-benzoquinone $+ 2H^+ + 2e \rightleftharpoons$ hydroquinone	0.6992
$Br_2(aq) + 2e \rightleftharpoons 2Br^-$	1.0874
$Ca^{2+} + 2e \rightleftharpoons Ca$	−2.84
$Cd^{2+} + 2e \rightleftharpoons Cd$	−0.4025
$Cd^{2+} + 2e \rightleftharpoons Cd(Hg)$	−0.3515
$Ce^{4+} + e \rightleftharpoons Ce^{3+}$	1.72
$Cl_2(g) + 2e \rightleftharpoons 2Cl^-$	1.3583
$HClO + H^+ + e \rightleftharpoons \frac{1}{2}Cl_2 + H_2O$	1.630
$Co^{2+} + 2e \rightleftharpoons Co$	−0.277
$Co^{3+} + e \rightleftharpoons Co^{2+}$	1.92
$Cr^{2+} + 2e \rightleftharpoons Cr$	−0.90
$Cr^{3+} + e \rightleftharpoons Cr^{2+}$	−0.424
$Cr_2O_7^{2-} + 14H^+ + 6e \rightleftharpoons 2Cr^{3+} + 7H_2O$	1.36
$Cu^+ + e \rightleftharpoons Cu$	0.520
$Cu^+ + 2CN^- + e \rightleftharpoons Cu(CN)_2^-$	1.12
$Cu^{2+} + e \rightleftharpoons Cu^+$	0.159
$Cu^{2+} + 2e \rightleftharpoons Cu$	0.340
$Cu^{2+} + 2e \rightleftharpoons Cu(Hg)$	0.345
$Eu^{3+} + e \rightleftharpoons Eu^{2+}$	−0.35
$1/2F_2 + H^+ + e \rightleftharpoons HF$	3.053
$Fe^{2+} + 2e \rightleftharpoons Fe$	−0.44
$Fe^{3+} + e \rightleftharpoons Fe^{2+}$	0.771
$Fe(CN)_6^{3-} + e \rightleftharpoons Fe(CN)_6^{4-}$	0.3610

(continued)

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TABLE C.1 *(continued)*

Reaction	Potential, V
$2H^+ + 2e \rightleftharpoons H_2$	0.0000
$2H_2O + 2e \rightleftharpoons H_2 + 2OH^-$	−0.828
$H_2O_2 + 2H^+ + 2e \rightleftharpoons 2H_2O$	1.763
$2Hg^{2+} + 2e \rightleftharpoons Hg_2^{2+}$	0.9110
$Hg_2^{2+} + 2e \rightleftharpoons 2Hg$	0.7960
$Hg_2Cl_2 + 2e \rightleftharpoons 2Hg + 2Cl^-$	0.26816
$Hg_2Cl_2 + 2e \rightleftharpoons 2Hg + 2Cl^-$ (sat'd. KCl)	0.2415
$HgO + H_2O + 2e \rightleftharpoons Hg + 2OH^-$	0.0977
$Hg_2SO_4 + 2e \rightleftharpoons 2Hg + SO_4^{2-}$	0.613
$I_2 + 2e \rightleftharpoons 2I^-$	0.5355
$I_3^- + 2e \rightleftharpoons 3I^-$	0.536
$K^+ + e \rightleftharpoons K$	−2.925
$Li^+ + e \rightleftharpoons Li$	−3.045
$Mg^{2+} + 2e \rightleftharpoons Mg$	−2.356
$Mn^{2+} + 2e \rightleftharpoons Mn$	−1.18
$Mn^{3+} + e \rightleftharpoons Mn^{2+}$	1.5
$MnO_2 + 4H^+ + 2e \rightleftharpoons Mn^{2+} + 2H_2O$	1.23
$MnO_4^- + 8H^+ + 5e \rightleftharpoons Mn^{2+} + 4H_2O$	1.51
$Na^+ + e \rightleftharpoons Na$	−2.714
$Ni^{2+} + 2e \rightleftharpoons Ni$	−0.257
$Ni(OH)_2 + 2e \rightleftharpoons Ni + 2OH^-$	−0.72
$O_3 + 2H^+ + 2e \rightleftharpoons H_2O_2$	0.695
$O_2 + 4H^+ + 4e \rightleftharpoons 2H_2O$	1.229
$O_2 + 2H_2O + 4e \rightleftharpoons 4OH^-$	0.401
$O_3 + 2H^+ + 2e \rightleftharpoons O_2 + H_2O$	2.075
$Pb^{2+} + 2e \rightleftharpoons Pb$	−0.1251
$Pb^{2+} + 2e \rightleftharpoons Pb(Hg)$	−0.1205
$PbO_2 + 4H^+ + 2e \rightleftharpoons Pb^{2+} + 2H_2O$	1.468
$PbO_2 + SO_4^{2-} + 4H^+ + 2e \rightleftharpoons PbSO_4 + 2H_2O$	1.698
$PbSO_4 + 2e \rightleftharpoons Pb + SO_4^{2-}$	−0.3505
$Pd^{2+} + 2e \rightleftharpoons Pd$	0.915
$Pt^{2+} + 2e \rightleftharpoons Pt$	1.188
$PtCl_4^{2-} + 2e \rightleftharpoons Pt + 4Cl^-$	0.758
$PtCl_6^{2-} + 2e \rightleftharpoons PtCl_4^{2-} + 2Cl^-$	0.726
$Ru(NH_3)_6^{3+} + e \rightleftharpoons Ru(NH_3)_6^{2+}$	0.10
$S + 2e \rightleftharpoons S^{2-}$	−0.447
$Sn^{2+} + 2e \rightleftharpoons Sn$	0.1375
$Sn^{4+} + 2e \rightleftharpoons Sn^{2+}$	0.15
$Tl^+ + e \rightleftharpoons Tl$	−0.3363
$Tl^+ + e \rightleftharpoons Tl(Hg)$	−0.3338
$Tl^{3+} + 2e \rightleftharpoons Tl^+$	1.25
$U^{3+} + 3e \rightleftharpoons U$	1.66
$U^{4+} + e \rightleftharpoons U^{3+}$	−0.52
$UO_2 + 4H^+ + e \rightleftharpoons U^{3+} + 2H_2O$	0.273
$UO_2^{2+} + e \rightleftharpoons UO_2$	0.163

(continued)

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TABLE C.1 *(continued)*

Reaction	Potential, V
$V^{2+} + 2e \rightleftharpoons V$	−1.13
$V^{3+} + e \rightleftharpoons V^{2+}$	−0.255
$VO^{2+} + 2H^+ + e \rightleftharpoons V^{3+} + H_2O$	0.337
$VO_2^+ + 2H^+ + e \rightleftharpoons VO^{2+} + H_2O$	1.00
$Zn^{2+} + 2e \rightleftharpoons Zn$	−0.7626
$ZnO_2^{2-} + 2H_2O + 2e \rightleftharpoons Zn + 4OH^-$	−1.285

[a] The data in this table are mainly taken from A. J. Bard, J. Jordan, and R. Parsons, Eds., "Standard Potentials in Aqueous Solutions," Marcel Dekker, New York, 1985 (prepared under the auspices of the Electrochemistry and Electroanalytical Chemistry Commissions of IUPAC). Other sources of standard potentials and thermodynamic data include: (1) A. J. Bard and H. Lund, Eds., "The Encyclopedia of the Electrochemistry of the Elements," Marcel Dekker, New York, 1973–1986. (2) G. Milazzo and S. Caroli, "Tables of Standard Electrode Potentials," Wiley-Interscience, New York, 1977. The data here are referred to the NHE based on a 1-atm standard state for H_2. See the footnote in Section 2.1.5 concerning the recent change in standard state.

Table C.2 Selected Formal Potentials in Aqueous Solution at 25°C in V *vs.* NHE[a]

Reaction	Conditions	Potential, V
$Cu(II) + e \rightleftharpoons Cu$	1 *M* NH_3 + 1 *M* NH_4^+	0.01
	1 *M* KBr	0.52
$Ce(IV) + e \rightleftharpoons Ce(III)$	1 *M* HNO_3	1.61
	1 *M* HCl	1.28
	1 *M* $HClO_4$	1.70
	1 *M* H_2SO_4	1.44
$Fe(III) + e \rightleftharpoons Fe(II)$	1 *M* HCl	0.70
	10 *M* HCl	0.53
	1 *M* $HClO_4$	0.735
	1 *M* H_2SO_4	0.68
	2 *M* H_3PO_4	0.46
$Fe(CN)_6^{3-} + e \rightleftharpoons Fe(CN)_6^{4-}$	0.1 *M* HCl	0.56
	1 *M* HCl	0.71
	1 *M* $HClO_4$	0.72
$Sn(IV) + 2e \rightleftharpoons Sn(II)$	1 *M* HCl	0.14

[a] The data in this table are taken mainly from G. Charlot, "Oxidation-Reduction Potentials," Pergamon, London, 1958. Additional values are found in J. J. Lingane, "Electroanalytical Chemistry," Interscience, New York, 1958, and L. Meites, Ed., "Handbook of Analytical Chemistry," McGraw-Hill, New York, 1963.

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